Multiplicities of eigenvalues

Definition (Algebraic Multiplicity).

For a matrix M, we say an eigenvalue λ of M has algebraic multiplicity k if λ is a root of the characteristic polynomial k-times.

Example. $\chi_\lambda(\lambda) = (1-\lambda)(\lambda+2)^2$.

$\Rightarrow$ $\lambda = 1$ has algebraic multiplicity one,
$\lambda = -2$ has algebraic multiplicity two.

If M is $n \times n$, then by the fundamental theorem of algebra,

$$n = \sum (\text{algebraic multiplicities of all eigenvalues})$$

Definition (Geometric Multiplicity).

For a matrix M, we say an eigenvalue λ of M has geometric multiplicity k if the λ-eigenspace has k parameters. That is, the solution to $(M - \lambda I) = \vec{0}$ had k free variables.

Example.

$$M = \begin{pmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{pmatrix},$$

for $\lambda = -2$ we see

$$M + 2I = \begin{pmatrix} 4 & 4 & 3 \\ -4 & -4 & -3 \\ 3 & 3 & 3 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 1 \\ 4 & 4 & 3 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\sim \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

Great are free variable (x_2) and $\lambda = -2$ eigenspace is

$$\left\{ t \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \;\middle|\; t \in \mathbb{R} \right\}$$

so, $\lambda = -2$ has geometric multiplicity one.

Proposition. If M is a square matrix with an eigenvalue λ, then

$$(\text{Geometric mult. of } \lambda) \leq (\text{Alg. mult. of } \lambda)$$

$\Rightarrow$ If M is $n \times n$, then

$$n \geq \sum (\text{geometric multiplicities of all eigenvalues})$$

Diagonalisable Matrices

Let M be an $n \times n$ matrix with eigenvalues $\lambda_1, \ldots, \lambda_n$ (listed with algebraic multiplicity). Let $\vec{v}_1, \ldots, \vec{v}_n$ be eigenvectors for $\lambda_1, \ldots, \lambda_n$ respectively.

Define $P = (\vec{v}_1 \ \vec{v}_2 \cdots \vec{v}_n)$ the $n \times n$ matrix with columns $\{\vec{v}_i\}$, $i = 1, \ldots, n$.

$$\begin{aligned} MP &= M(\vec{v}_1 \ \vec{v}_2 \cdots \vec{v}_n) = (M\vec{v}_1 \ \cdots \ M\vec{v}_n) \\ &= (\lambda_1 \vec{v}_1 \cdots \lambda_n \vec{v}_n) \\ &= P \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}. \end{aligned}$$

So set $D = \begin{pmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{pmatrix}$, we get

$$MP = PD.$$

If P is invertible, $M = PDP^{-1}$. If this is possible, then we say M has been diagonalised.

<u>Definition (Diagonalisable Matrix)</u>.

For M, an $n \times n$ matrix, we say M is diagonalisable if there exists an invertible matrix P and diagonal matrix D such that

$$M = PDP^{-1}.$$

If so, the diagonal entries of D are the eigenvalues of M and the columns of P are the corresponding eigenvectors of M.

<u>Example</u>. $M = \begin{pmatrix} 4 & 4 \\ 2 & 1 \end{pmatrix} \in Mat_2(\mathbb{Z}_5)$, let

$$D = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}, \quad P = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix}.$$

P is invertible, so

$$M = PDP^{-1} = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 4 & 1 \\ 2 & 4 \end{pmatrix}.$$

Theorem. A matrix M is diagonalisable iff for all eigenvalues λ,

$$(\text{alg. mult. of } \lambda) = (\text{geo. mult. of } \lambda).$$

In this case, we construct P by taking columns corresponding to "parameters in the eigenspaces".

Example. $M = \begin{pmatrix} 4 & 4 \\ 2 & 1 \end{pmatrix} \in Mat_2(\mathbb{Q})$. Find M^k, $k \geq 1$.

$$M = PDP^{-1} \Rightarrow M^2 = PDP^{-1}PDP^{-1}$$
$$= PD^2P^{-1}$$

$$\Rightarrow M^3 = PD^2P^{-1}PDP^{-1}$$
$$= PD^3P^{-1}$$

$$\Rightarrow M^k = PD^kP^{-1} = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 2^k & 0 \\ 0 & 3^k \end{pmatrix} \begin{pmatrix} 4 & 1 \\ 2 & 4 \end{pmatrix}$$

Cayley-Hamilton Theorem

Definition (Matrix Polynomials).

Let $M \in Mat_n(F)$, $p(x) = \sum_{k=0}^{n} a_k x^k$ be a polynomial in a variable x with coefficients $a_0, a_1, \ldots, a_n \in F$. The matrix polynomial $p(M)$ is given by

$$p(M) = a_0 I + a_1 M + a_2 M^2 + \cdots + a_n M^n.$$

Theorem (Cayley-Hamilton Theorem).

Let $M \in Mat_n(F)$ with characteristic polynomial $\chi_M(\lambda) = \det(M - \lambda I)$. Then,

$$\chi_M(M) = \begin{pmatrix} 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{pmatrix}.$$

Proof (for a nice case: diagonalizable M).

Step 1: Consider $D = \begin{pmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{pmatrix}$.

Then notice

$$D^k = \begin{pmatrix} \lambda_1^k & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n^k \end{pmatrix}, \quad aD = \begin{pmatrix} a\lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & a\lambda_n \end{pmatrix}.$$

Then for any polynomial $p(x)$,

$$p(D) = \begin{pmatrix} p(\lambda_1) & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & p(\lambda_n) \end{pmatrix}.$$

Now for the characteristic polynomial $\chi_D(\lambda)$, $\lambda_1, \ldots, \lambda_n$ are eigenvalues of D, so

$$\chi_D(\lambda_i) = 0 \in F, \quad i \in \{1, \ldots, n\}.$$

$$\Rightarrow \chi_D(D) = \begin{pmatrix} \chi_D(\lambda_1) & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \chi_D(\lambda_n) \end{pmatrix}$$

$$= \begin{pmatrix} 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{pmatrix}$$

Step 2: Suppose $M = PDP^{-1}$ diagonalisable.
For any polynomial $p(x) = \sum_{k=0}^{n} a_k x^k$,

$$
\begin{aligned}
p(M) &= a_0 I + a_1 M + a_2 M^2 + \cdots + a_n M^n \\
&= a_0 I + a_1 PDP^{-1} + a_2 PD^2P^{-1} + \cdots + a_n PD^nP^{-1} \\
&= P(a_0 I + a_1 D + a_2 D^2 + \cdots + a_n D^n)P^{-1} \\
&= P\,p(D)\,P^{-1}
\end{aligned}
$$

Therefore, $\chi_M(M) = P\chi_M(D)P^{-1}$.

$$
\begin{aligned}
\det(M - \lambda I) &= \det(PDP^{-1} - \lambda I) \\
&= \det(PDP^{-1} - \lambda(PP^{-1})) \\
&= \det(P(D - \lambda I)P^{-1}) \\
&= \det P \det(D - \lambda I) \det(P^{-1}) \\
&= \det(D - \lambda I).
\end{aligned}
$$

$\Rightarrow \chi_M(\lambda) = \chi_D(\lambda)$

$$
\begin{aligned}
\Rightarrow \chi_M(M) = P\chi_M(D)P^{-1} &= P\chi_D(D)P^{-1} \\
&= P\begin{pmatrix} 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{pmatrix}P^{-1} \\
&= \begin{pmatrix} 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{pmatrix}
\end{aligned}
$$

Application: Computing Inverses.

Example. Let $M = \begin{pmatrix} 3 & -1 \\ 2 & 1 \end{pmatrix} \in Mat_2(\mathbb{R})$.

$$\chi_M(\lambda) = \lambda^2 - 4\lambda + 5 .$$

By Cayley-Hamilton Theorem,

$$\chi_M(M) = M^2 - 4M + 5I = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\Rightarrow M^2 - 4M = -5I$$
$$\Rightarrow M(M - 4I) = -5I$$
$$\Rightarrow -\tfrac{1}{5}M(M-4I) = I$$
$$\therefore M^{-1} = -\tfrac{1}{5}(M - 4I).$$

Theorem. If M is invertible, then there is some polynomial $p(x)$ such that

$$M^{-1} = p(M)$$

Example. $\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \in Mat_3(\mathbb{Z}_3)$. Can compute $\chi_M(\lambda) = 2\lambda^3 + 2$. Then

$$2M^3 + 2I = 0$$

$$\Rightarrow 2M^3 = I$$

$$\Rightarrow M(2M^2) = I$$

$$\Rightarrow M^{-1} = 2M^2 = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix}$$

Take $p(x) = 2x^2$.